

Technology Stack

Requirement Analysis • AI/ML and GenAI Track

Layer	Technology	Purpose
Language	Python 3.12	Backend logic
Web framework	FastAPI + Uvicorn	Routing, validation, auto API docs at /docs
Templating	Jinja2	Server-rendered pages (index, result, all_users, error)
Validation	Pydantic	Typed input models (UserInput, FeedbackRequest)
Database	SQLite via SQLAlchemy 2.x	Persist users and plans
Generative AI	Google Gemini via the google-genai SDK (model set by environment variable, default gemini-2.5-flash)	Workout plan, plan revision, nutrition tip
Frontend	HTML, CSS, small vanilla JavaScript	Form, styling, loading overlay
Configuration	python-dotenv and a .env file	API key, admin credentials, model names, database URL
Deployment	Docker (python:3.12-slim)	Containerised run with a /data volume
Version control	Git and GitHub	Source management

Why this stack

- FastAPI: fast to build, built-in validation and interactive docs.
- SQLite: zero setup, ideal for a prototype; can be swapped for PostgreSQL through DATABASE_URL.
- Gemini: strong text generation with a simple SDK and a fast, low-cost "flash" model.